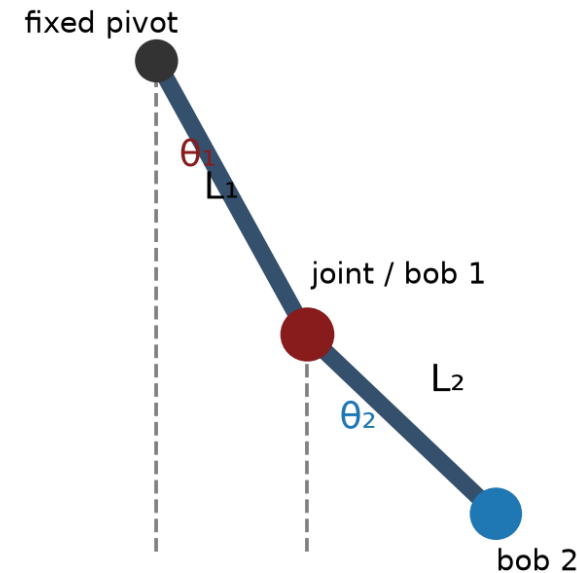


Double pendulum: from video to physics

- Lecture 04 · Monday, October 5, 2026
- Lab 3 takes place October 19-23; October 12 has no class or lab.
- Today: understand the device, predict motion, prepare measurements and analysis.



Angles are absolute from downward vertical; θ_2 is not $\theta_2 - \theta_1$.

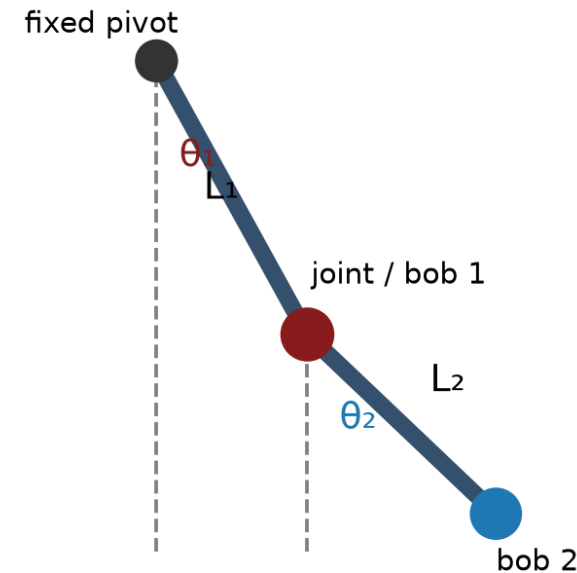
What will students actually do?

- Build and measure a two-link pendulum; record four or more useful releases.
- Track two moving bob centers with the high-speed camera and MATLAB app.
- Convert saved coordinates to two angles; compare traces and spectra with a model.
- Keep raw MP4, MAT, masses, lengths, settings, and a trial log.



The two links and two angles

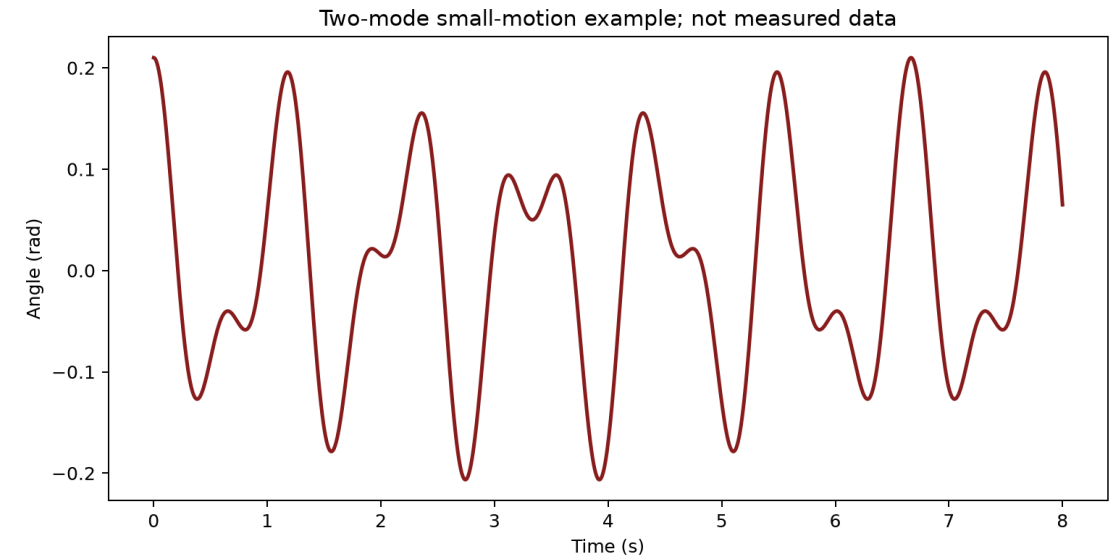
- Upper link: fixed pivot to joint/bob 1; lower link: joint to bob 2.
- θ_1 and θ_2 are each measured from the downward vertical, positive to the right.
- θ_2 is the absolute orientation of link 2; $\theta_2 - \theta_1$ is its relative angle.
- Measure pivot-to-joint L_1 and joint-to-bob-center L_2 .



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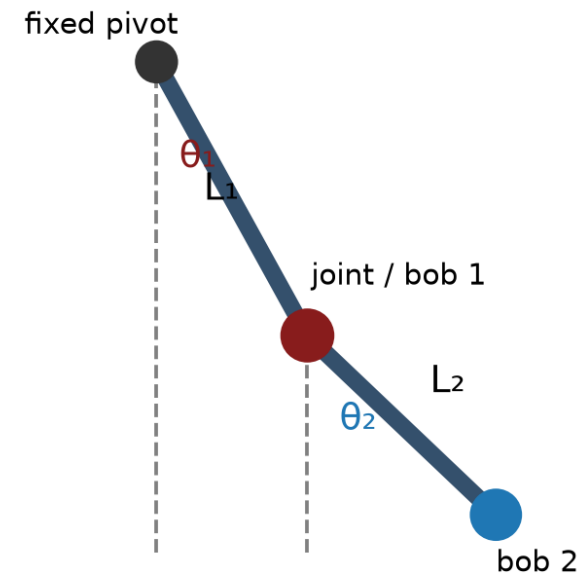
One pendulum versus two

- Lab 2 had one angle and one principal oscillation frequency.
- Lab 3 has two angles: motion of one link changes the other.
- Energy transfers between the links; even small motion can contain two normal modes.
- At large amplitudes the sine and angle-difference terms make the motion nonlinear.



The physical model we will test

- m_1, m_2 : upper/lower uniform bar masses; M_1, M_2 : lumped joint/lower-bob masses.
- L_1, L_2 : pivot-to-joint and joint-to-lower-bob-center distances.
- Assume planar rigid links, frictionless pivots, and release from rest for baseline.
- Measured masses and lengths replace last year's illustrative numbers.



Angles are absolute from downward vertical; θ_2 is not $\theta_2 - \theta_1$.

Where the restoring torque comes from

- Gravity acts at each bar's center and at each bob/hardware mass.
- For a displaced angle θ , the horizontal lever arm is distance $\times \sin \theta$.
- Bar centers lie at $L_1/2$ and $L_2/2$ for uniform bars.
- Potential energy rises as each mass moves above its hanging position.

Energy of the double pendulum

$$T = \frac{1}{2}I_{11}\dot{\theta}_1^2 + \frac{1}{2}I_{22}\dot{\theta}_2^2 + C\cos(\theta_1 - \theta_2)\dot{\theta}_1\dot{\theta}_2$$

$$V = G_1(1 - \cos \theta_1) + G_2(1 - \cos \theta_2)$$

$$I_{11} = (m_1/3 + M_1 + m_2 + M_2)L_1^2, \quad I_{22} = (m_2/3 + M_2)L_2^2$$

$$C = (m_2/2 + M_2)L_1L_2$$

$$G_1 = (m_1/2 + M_1 + m_2 + M_2)gL_1, \quad G_2 = (m_2/2 + M_2)gL_2$$

Equations used by the supplied simulation

The coupled accelerations are solved together at each time step:

$$\Delta = \theta_1 - \theta_2$$

$$I_{11}\ddot{\theta}_1 + C\cos\Delta\ddot{\theta}_2 = -C\sin\Delta\dot{\theta}_2^2 - G_1\sin\theta_1$$

$$C\cos\Delta\ddot{\theta}_1 + I_{22}\ddot{\theta}_2 = C\sin\Delta\dot{\theta}_1^2 - G_2\sin\theta_2$$

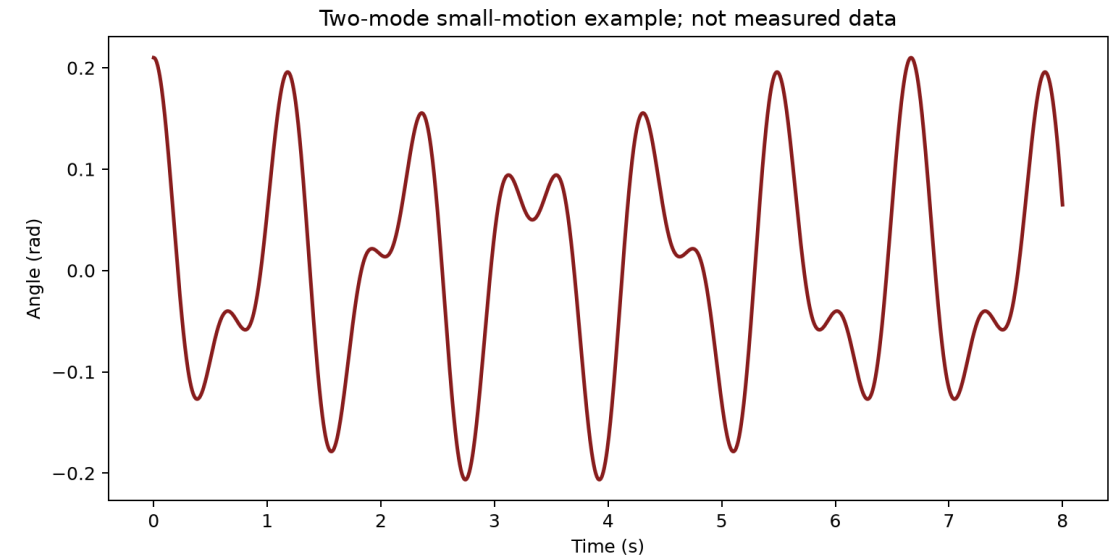
MATLAB solves this 2×2 system inside ode45; students do not write an ODE solver.

Why the coupling terms matter

- The off-diagonal $C \cos\Delta$ couples both angular accelerations.
- The $\sin\Delta \times$ angular-speed-squared terms vary as the geometry changes.
- Setting both bar masses to zero gives the ideal point-mass limit.
- A real bar has mass: leaving it out can shift predicted timing.

Small motion: two normal modes

- For small angles, $\sin\theta \approx \theta$ and $\cos(\theta_1 - \theta_2) \approx 1$.
- The linearized model is $M_0 \ddot{q} + Kq = 0$, with $q = [\theta_1, \theta_2]^T$.
- In one mode links move mostly together; in the other they move mainly oppositely.
- The two frequencies depend on measured masses and lengths.



Large motion and initial conditions

- At large angles, the small-angle approximation is no longer reliable.
- Two nearly identical starts may separate over time as coupling changes.
- A short finite trace shows sensitivity; it does not by itself prove chaos.
- Release without a push; document both initial angles and whether initial speeds are near zero.

Design the four required trials

- A: small nonzero, roughly aligned links, released from rest.
- B-D: at least three larger initial-angle combinations.
- Include one pair with similar but distinct starting angles to compare sensitivity.
- Record actual $\theta_1(0)$, $\theta_2(0)$, masses, lengths, frame rate, and file names.

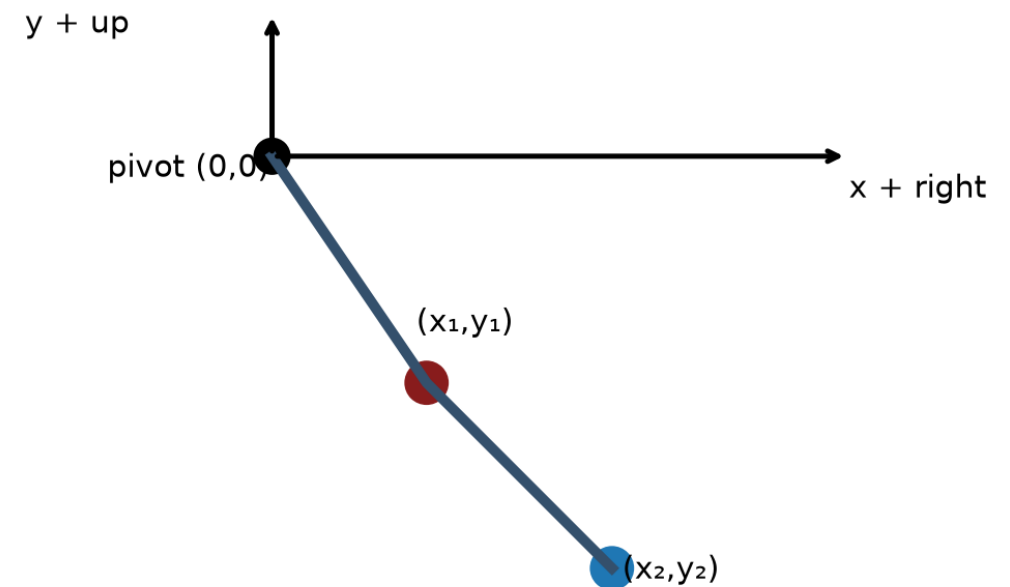
What the high-speed camera records

- The camera stores images, not angles or angular velocities.
- Use black background, bright bobs, sharp focus, and full-motion framing.
- Set 1280×1024 and nominal 1000 capture frames/s, then record actual setting.
- Saved MP4 playback frame rate may differ from capture frame rate.



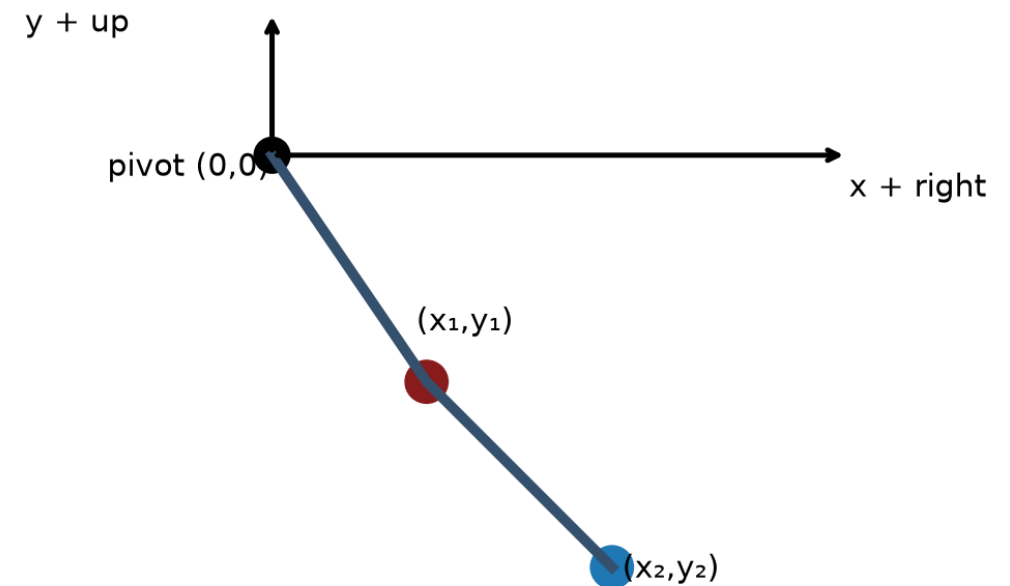
From video to two trajectories

- Select Double Pendulum in Pendulum Motion Analyzer V4.
- Enter capture frame rate and known calibration distance between black marks.
- Locate fixed pivot and centers of both bobs in first frame.
- Quick Review first; rerun setup if >10% are missing, then analyze all frames.



What the MAT file contains

- ExpData: time [s], x_1 [mm], y_1 [mm], x_2 [mm], y_2 [mm].
- App coordinates are relative to pivot: x is right-positive, y is up-positive.
- The lower-link vector is $(x_2 - x_1, y_2 - y_1)$.
- Inspect one saved file before leaving; preserve the raw MP4 as well.



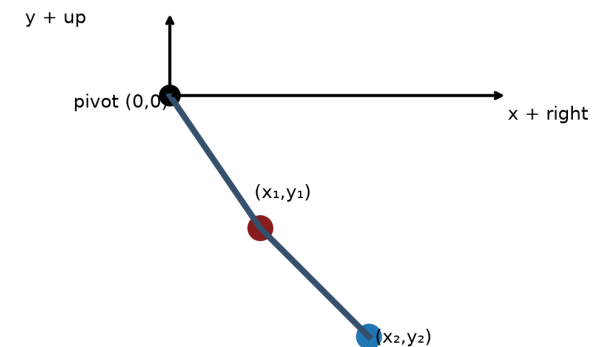
Convert tracked positions to angles

From the app's pivot-relative, right-positive and up-positive coordinates:

$$\theta_1 = \text{unwrap}(\text{atan2}(x_1, -y_1))$$

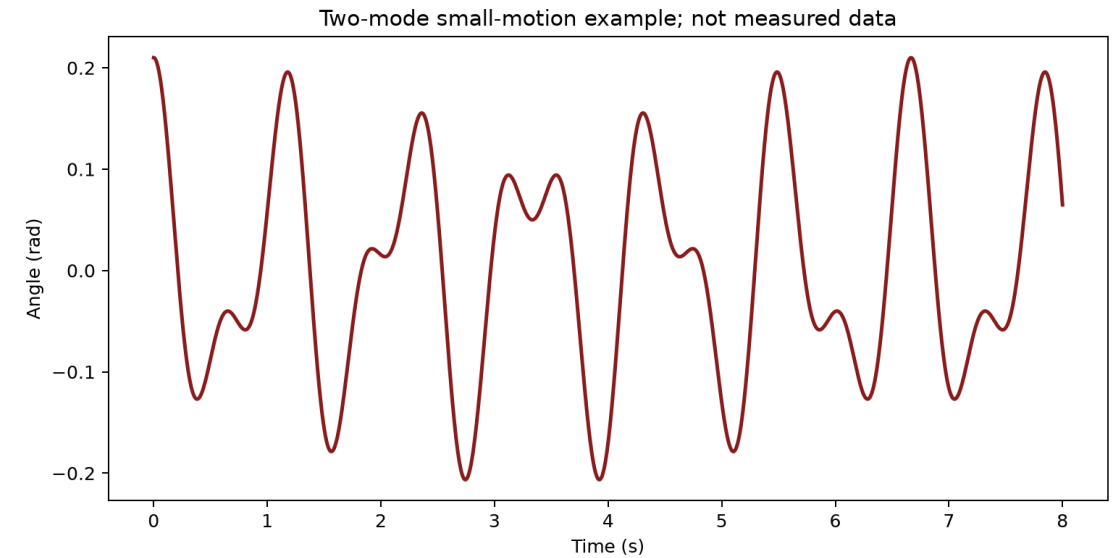
$$\theta_2 = \text{unwrap}(\text{atan2}(x_2 - x_1, -(y_2 - y_1)))$$

Both angles should be near zero when both links hang down.
Check one known frame and sign before processing all trials.



Velocity and frequency: what can we trust?

- Angular velocity is the slope of the angle trace; missing points and jitter can amplify noise.
- FFT displays frequencies present in a finite record; subtract the mean first.
- A coupled pendulum need not have a single “pendulum frequency.”
- Use spectra to compare dominant components; inspect time traces and record duration too.



Compare measured and simulated motion

- Use measured masses, lengths, initial angles, and approximate initial speeds.
- Overlay $\theta_1(t)$ and $\theta_2(t)$ for each trial; compare early timing and shape.
- For large-angle trials, focus on short-time agreement and qualitative behavior.
- Explain discrepancies: release push, calibration, friction, air drag, out-of-plane motion.

What model omits

- The required baseline model has no damping. Real pivots and moving links lose energy.
- Amplitude decay is observable, but one recording cannot isolate air drag from pivot friction.
- Joint hardware is treated as a lumped mass; real bars and hardware are not perfectly uniform.
- Do not tune damping to make one plot look good without documenting a new model.

Pre-Lab 3: different from last year

- No student coding: run supplied Colab cells and answer in Word/PDF.
- Compare a low-angle release with two near-identical large-angle starts.
- Predict which trace is simpler; quantify divergence and inspect both angle spectra.
- Explain why model mismatch does not automatically mean a tracking error.

In the lab: a practical sequence

- 1 Measure parts; log which hardware belongs to which link/bob.
- 2 Set camera, calibrate, frame, focus, and record four releases.
- 3 Save MP4; in app select Double Pendulum and Quick Review.
- 4 Analyze full video, save MAT, make trial plots; show one complete set to GTA.



Before leaving: evidence checklist

- Four trial labels with measured initial conditions.
- Masses and lengths with units; capture fps and calibration distance.
- Raw MP4 and ExpData MAT for each trial.
- For one trial: two angle traces, two angular-velocity traces, and spectra checked with GTA.

Question for discussion

- If θ_1 and θ_2 start almost equal, must they remain equal?
- What extra observation distinguishes a bad tracking trace from real coupling?
- Which parameter in the model changes if you add washers to the lower bob?
- Use the pre-lab to test your answer before Lab 3.